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Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2022/2023

Course code & title: MTH 111 (General Mathematics I) Duration: 1HR: 30MINS

REG NO:

INSTRUCTION: ANSWER ALL QUESTIONS. CIRCLE THE CORRECT OPTION IN THIS QUESTION PAPER

TYPE: COET

NAME: MATRIC NO: DEPARTMENT:

1. Express $4\pi/6$ in degrees. (a) 120° (b) 90° (c) 150° (a) 60°
2. What is the root of the quadratic equation $X^2 - 5x + 6 = 0$? (a) $x = 2$ or 3 (b) $x=3$ or 2 (c) $x=4$ or 5 (d) $x=5$ or 6
3. If the roots of a quadratic equation are $x = 1$ and $x = 1$, what is the equation? (a) $x^2 - 2x + 1 = 0$ (b) $x^2 - 3x + 2 = 0$ (c) $x^2 - 5x + 1 = 0$ (d) $x^2 - 2x + 2$
4. What type of quadratic root is obtained If the discriminant $\sqrt{b^2 - 4ac} = 0$? (a) Complex root (b) Real and Equal root (c) Real and Distinct root (d) Negative root
5. Evaluate $\frac{4!}{(4-2)!2!}$? @12 (b) 6 (c) 4 (d) 3
6. Solve for x in the equation $5x + 6 = 21$. (a) 15 (b) 5 (c) 3 (d) 2
7. A complex root is obtained from a quadratic equation when its discriminant is? (a) $\sqrt{b^2 - 4ac} = 0$ (b) $\sqrt{b^2 - 4ac} \geq 0$ (c) $\sqrt{b^2 - 4ac} < 0$ (d) $\sqrt{b^2 - 4ac} > 0$
8. What is the sum and product of the roots of the quadratic equation $x^2 - 4x - 3 = 0$? (a) $\alpha + \beta = 1$ and $\alpha\beta = 5$ (b) $\alpha + \beta = 2$ and $\alpha\beta = 5$ (c) $\alpha + \beta = 3$ and $\alpha\beta = 4$ (d) $\alpha + \beta = 4$ and $\alpha\beta = -3$
9. Expand $(x+2)^3$ using the binomial expansion process (a) $x^3 + 12x^2 + 12x + 8 = 0$ (b) $x^3 + 6x^2 + 8x + 8$ (c) $4x^2 + 12x + 8 = 0$ (d) $x^3 + 6x^2 + 12x + 8 = 0$
10. Find two numbers whose difference is 10 and whose product is 200? @ -20 and 10 (b) 10 and 2 (c) 30 and -20 (d) 100 and -2
11. In a class of 45 students, 30 offers Mathematics, 20 offers Chemistry. How many students offers both? (a) 1 (b) 3 (c) 5 (d) 7
12. What law qualifies $2 \times (3 \times 4) = (2 \times 3) \times 4$? (a) Identity law (b) Associative law (c) Existence law (d) Commutative law
13. $a * c = e * a = a$ Shows (a) inverse (b) Associativity

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- (c) Closure (d) Identity
14. The sum of the sequence $\sum_{x=1}^4 4x \dots$ is (a) 40 (b) 30 (c) 55 (d) 35
15. Compute the sum of $\sum_{T=1}^4 T^2 - 2T + 2$ (a) 18 (b) 26 (c) 30 (d) 16
16. What is the common difference between the set 1, 4, 7, 10, 13, 16? (a) 4 (b) 5 (c) 3 (d) 1
17. The n th of an arithmetic sequence is given as..... (a) $T_n = a^{r-1}$ (b) $T_n = 2a^{r-1}$
(c) $T_n = \frac{a^{r-1}}{s}$ (d) $T_n = a + (n-1)d$
18. The 5th term of AP is expressed as.....? (a) $T_n = a + (n-1)d$ (b) $T_n = a + (n-3)d$
(c) $T_n = a + (3n-1)d$ (d) $T_n = a + 4d$
19. What is the 13th term of the arithmetic sequence 2, 6, 10, 14, 18, (a) 50 (b) 43 (c) 44 (d) 34
20. The 4th term of a geometric progression is 40 and its 1st is 5. Determine its 7th (a) 540 (b) 640
(c) 450 (d) 320
21. The n th sum of an arithmetic progression is given as..... (a) $S_n = \frac{n}{2}[2a + (n-1)d]$ (b) $S_n = \frac{n}{2}[2a + d]$
(c) $S_n = \frac{n}{2}[2a + 5d]$ (d) $S_n = \frac{n}{2}[2a + 10d]$
22. Find the sum of the first 10 terms of $2+5+8+11+\dots$ (a) 55 (b) 543 (c) 55 (d) 342
23. What is the common ratio of the set 3, 9, 27, 81..... (a) 3 (b) 9 (c) 12 (d) 49
24. In the sum of geometric progression, if $r > 1$, what the sum of G.P. becomes.....
(a) $S_n = \frac{a(r^n-1)}{r-1}$ (b) $S_n = \frac{a(-1-r^n)}{r-1}$ (c) $S_n = \frac{a(1-r^n)}{r-1}$ (d) $S_n = \frac{a(-1-r^n)}{r-1}$
25. Solve the equation $2x^2 - 10 = 0$ (a) $x = \pm\sqrt{5}$ (b) $x = \pm\sqrt{10}$ (c) $x = \pm\sqrt{13}$ (d) $x = \pm\sqrt{15}$
26. Evaluate $\binom{10}{7}$ (a) 100 (b) 230 (c) 120 (d) 410
27. If Z represents the set of integers, then $\{y: -3 < y < 2\}$ is (a) $\{-3, 2\}$ (b) $\{0, 1\}$ (c) $\{-3, -2, -1, 0, 1, 2\}$
(d) $\{-2, -1, 0, 1\}$
28. The nonnegative integer is equivalent to (a) $\{x: x \geq 0\}$ (b) $\{x \in \mathbb{Z}: 0 < x < 1\}$ (c) 0 (d) None
29. Given that $360 = 2\pi \text{ rad}$, what is the value of a rad? (a) 45 (b) 57 (c) 38 (d) 47
30. Resolve $\frac{3x+2}{(x+2)(x-5)}$ into partial fraction (a) $\frac{3}{2(x+1)} + \frac{5}{(2x+1)}$ (b) $9 - \frac{5}{7(2x-1)}$ (c) $\frac{4}{7(x+2)} + \frac{17}{7(x-5)}$ (d) undefined

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31. Using the method of comparing coefficient, resolve $\frac{7x+1}{(3x-1)(2x+3)}$ into partial fraction. (a) $\frac{4}{(3x-1)} + \frac{23}{(2x+3)}$ (b) $\frac{2}{(3x-1)} + \frac{5}{(2x+3)}$ (c) $\frac{34}{(3x-1)} + \frac{23}{(2x+3)}$ (d) $\frac{11}{(3x-1)} + \frac{3}{(2x+3)}$
32. Given that set $A = \{-2, -1, 0, 1, 2, 3\}$ and $B = \{4, 5, \dots\}$, what is $A - B$? (a) $A - B = \{6, 2, \dots\}$ (b) $A - B = \{-2, -1, 0, 1, 2, 3\}$ (c) $A - B = \{\dots, 3, 2, 1\}$ (d) $A - B = \{6, 2, 0, -2, \dots\}$
33. If the roots of $2x^2 - x - 2 = 0$ are α and β . Find $\alpha + \beta$ and $\alpha\beta$ (a) $\frac{1}{2}$ and 1 (b) $-\frac{1}{2}$ and 2 (c) $\frac{1}{2}$ and -1 (d) $\frac{1}{2}$ and $\frac{1}{2}$
34. For what values of x is the quadratic equation $x^2 - 7x + 12 = 0$ (a) 2 and 5 (b) 12 and 7 (c) 2 and 7 (d) 3 and 4
35. The second term of the arithmetic progression is (a) $T_2 = a + 4d$ (b) $T_2 = a + 3d$ (c) $T_2 = a + 2d$ (d) $T_2 = a + d$
36. The sum of an infinite geometry progression is express as. (a) $S = \frac{a}{1-r}$ (b) $S = \frac{a}{r+1}$ (c) $S = \frac{a}{1-2r}$ (d) $S = \frac{a}{2r-1}$
37. What is the quadratic equation given that 3 and 4 are its roots? (a) $x^2 - 7x + 12$ (b) $x^2 - 7x + 2$ (c) $x^2 - 4x + 12$ (d) $2x^2 - 7x + 12$
38. State De Morgan's laws. (a) $(A \cup B)^c = A^c \cap B^c$ & $(A \cap B)^c = A^c \cup B^c$ (b) $(A \cup B)^c = A \cup B$ & $(A \cap B)^c = A \cup B$ (c) $(A \cup B)^c = A \cup B$ & $(A \cap B)^c = A \cup B$ (d) $(A \cup B)^c = A \cap B$ & $(A \cap B)^c = A \cap B$
39. If $U = \{x; 2 \leq x \leq 12\}$ and $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{2, 3, 5, 7, 11\}$ and $C = \{3, 5, 7, 9, 11\}$. Find; $B^c \cap A^c$ (a) $B^c \cap A^c = \{1, 2, 3\}$ (b) $B^c \cap A^c = \{9\}$ (c) $B^c \cap A^c = \{2, 8\}$ (d) $B^c \cap A^c = \{7, 8, 9\}$
40. In a class of 120 students, 60 speaks French, 55 speak English and 5 speak both French and English. How many students neither speak French nor English? (a) 30 (b) 10 (c) 55 (d) None
41. Which of the following is irrational number? (a) 3 (b) π (c) $\sqrt[3]{3}$ (d) $\frac{3}{4}$
42. If a complex number $r = a + ib$ then what is $1/r$. (a) $\sqrt{a^2 + b^2}$ (b) $\sqrt{a^3 + b^3}$ (c) $\sqrt{a^2 + b^2}$ (d) $\sqrt{a^2}$
43. If a complex number $r = a + ib$ then its conjugate will be (a) $a + ib$ (b) $a - ib$ (c) a (d) ib
44. Express the imaginary complex part of i^3 (a) i (b) i^2 (c) $\sqrt{-1}$ (d) 0
45. Simplify $(2 + 5i) + (3 + 4i)$ (a) $2 + i$ (b) $5 + 9i$ (c) $-6i$ (d) $2 + 7i$
46. Simplify $(2 - 5i) + (3 + 4i)$ (a) $5 - i$ (b) $5 + 9i$ (c) $-6i$ (d) 0

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47. Simplify $(2 + 5i)(2 - 5i)$ (a) 25 (b) 15 (c) 20 (d) 29
48. The 4th term of a GP is 40 and its first term is 5. Find the 7 term (a) 128 (b) 140 (c) 100 (d) 640
49. The AP sequence 3, 7, 11, has the n th term of 159, Find n (a) 10 (b) 20 (c) 30 (d) 40
50. The 3rd term of a GP is 32 and the 6th term is 4, what is the common ratio? (a) $r = \frac{1}{2}$ (b) $r = \frac{3}{4}$ (c) $\frac{3}{2}$ (d) $r = 1$

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UNIVERSITY OF FAISALABAD
FIRST SEMESTER EXAMINATION 2019/2020
Course code & title: MTH 111 (General Mathematics I)

Duration: 2 ¼ hrs

TYPE F

Instruction: Answer all questions. NOTA stands for none of the above

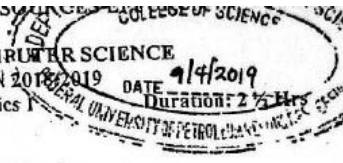
- Given the universal set $U = \{1, 2, 3, \dots, 10\}$, $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 5, 7, 9, 10\}$. Find $A \cup (B \cap C)$ (a) $\{3, 5, 7\}$ (b) $\{3, 5, 7, 9, 10\}$ (c) $\{1, 2, 3, \dots, 10\}$ (d) $\{2, 4, 5, 6, 7, 8, 9, 10\}$
- Given $P = \{1, 3, 5, 7, 9, 11\}$ and $Q = \{2, 4, 6, 8, 10, 12\}$. Determine the relationship between P and Q (a) $P \subset Q$ (b) $Q \subset P$ (c) $P = Q$ (d) $P \cap Q = \emptyset$
- If $A = \{(x, y) : y^2 = 4x\}$ and $B = \{(x, y) : y^x = 16\}$, find the elements of $A \cap B$ (a) $\{4, 16\}$ (b) $\{2, 4\}$ (c) $\{8, 4\}$ (d) $\{2, 8\}$
- The set $A = \{x, x \in \mathbb{N}, \text{ and } x^2 - 3x + 2 = 0\}$ is (a). Null set (b). Finite set (c). Infinite set (d). NOTA
- The set $A = \{x, x \in \mathbb{R}, \text{ and } x^2 = 9, 2x = 4\}$ is (a). Empty set (b). Singleton set (c). Infinite set (d). NOTA
- Let $A = \{x : x \text{ is a letter in the word FOLLOW}\}$, $B = \{y : y \text{ is a letter in the word WOLF}\}$, then (a). A & B are disjoint (b). $A = B$ (c). $A - B$ (d). NOTA
- Find the cardinal number of the set $A = \{x : x \in \mathbb{Z}, 2 < x < 7\}$ (a). 4 (b). 5 (c). 6 (d). 7
- If $A = \{5, 7, 8, 9\}$, $B = \{3, 4, 5, 6\}$ and $C = \{2, 4, 6, 8, 10\}$; where n is total number of distinct elements in a set. Find $n(A) + n(B) - n(C)$ (a). 1 (b). 2 (c). 3 (d). 4
- $A^c - B^c$ is equal to: (a). $B - A$ (b). $A - B$ (c). $A = B$ (d). NOTA
- The first term of AP is 4 and its last term is 73. Find the number of terms if the sum of the sequence is 924 (a) 24 (b). 21 (c). 73 (d). 60
- Find the sum of the first 10 terms of $\frac{1}{4}, \frac{1}{2}, \dots$ (a) 100.5 (b). 250.5 (c). 255.5 (d). 120.5
- If the sum of the first 5 terms of a GP is 25 and $r = 3$, find the first term (a) 0.1 (b). 0.3 (c). 0.4 (d). 0.2
- Solve the linear equation $\frac{1}{5}(10x - 20) = x - 3$ (a) $\{-1\}$ (b). $\{17\}$ (c). $\{3\}$ (d). $\{1\}$
- Find $(1 - 3x)(x^2 + 2x - 5)$ (a) $-3x^3 - 5x^2 + 17x - 5$ (b). $3x^2 - x - 4$ (c). $-3x^3 + 7x^2 - 5$ (d). NOTA
- What is the solution of $(2x - 1)^2 = 9$? (a) $\{-1\}$ (b). $\{2\}$ (c). $\{-1, 2\}$ (d). no solution
- Resolve $\frac{2}{x-3} + \frac{4}{5x+2}$ (a) $\frac{6}{6x-1}$ (b). $\frac{6}{(x-3)(5x+2)}$ (c). $\frac{6x+16}{(x-3)(5x+2)}$ (d). $\frac{14x-8}{(x-3)(5x+2)}$
- Compute i^{42} (a) 1 (b). $\sqrt{-1}$ (c). -1 (d). i
- Add or subtract $(5+7i) + (3-4i) + (6-3i)$ (a) $-2-5i$ (b). $3-4i$ (c). $5+7i$ (d). $2+6i$
- Multiply $(4-5i)$ and $(3+2i)$ (a) $22-7i$ (b). $3+2i$ (c). $30+7i$ (d). $20-6i$
- Multiply $(5+8i)$ and $(5-8i)$ (a) -39 (b). $25+64i$ (c). 89 (d). 39
- Find the numerator of the quotient $\frac{5-3i}{2+7i}$ in standard form (a). $-11-41i$ (b). $31+21i$ (c). $16-30i$ (d). $11+41i$
- Given $\frac{7-4i}{4+3i}$ find the denominator (a). 25 (b). 20 (c). 28 (d). undefined
- Find the sum or difference of $(3-5i) - (-5+11i) + (9+6i)$ in its standard form (a). $7+12i$ (b). $7-10i$ (c). $17+12i$ (d). $17-10i$
- Write $\frac{-24+\sqrt{-126}}{3}$ in the form $a+bi$ (a). $-8+14i$ (b). $-8-i8$ (c). $-8+i3.74$ (d) NOTA
- The sum of $(2+3i)$ and $(-3-5i)$ is in what quadrant? (a). 1st (b). 2nd (c). 3rd (d). 4th
- The difference of $(4+2i)$ and $(-2+3i)$ is in what quadrant? (a). 1st (b). 2nd (c). 3rd (d). 4th
- Find the difference of $(-1-3i)$ and $(-2+4i)$ (a). $1-7i$ (b). $-1+7i$ (c). $-3-7i$ (d). $3+7i$
- Given $Z = 4+3i$, find the modulus and argument of Z a. $(5, 36^\circ 52')$ b. $(4, 36^\circ)$ c. $(3, 45^\circ)$ d. $(7, 36^\circ 52')$
- Partial fractions of $\frac{x^2+1}{x^3+1}$ will be of the form (a) $\frac{A}{x-1} - \frac{B}{x^2-x+1}$ (b) $\frac{A}{x+1} - \frac{B}{x^2-x+1}$ (c) $\frac{A}{x+1} - \frac{Bx+C}{x^2-x+1}$ (d) NOTA
- Resolve $\frac{x}{(x+2)(x-3)}$ (a) $\frac{2}{5(x+2)} - \frac{3}{5(x-3)}$ (b) $\frac{2}{5(x+2)} + \frac{3}{5(x-3)}$ (c) $\frac{2}{5(x-2)} + \frac{3}{5(x-3)}$ (d) $\frac{3}{5(x-3)} - \frac{2}{5(x+2)}$
- $\frac{1}{(x-3)(x-2)}$ is equal to (a) $\frac{1}{x^2} - \frac{1}{x-2}$ (b) $\frac{1}{x-3} - \frac{1}{x-2}$ (c) $\frac{1}{x+3} - \frac{1}{x+2}$ (d) $\frac{1}{x^2} - \frac{1}{x-2}$
- Partial fractions of $\frac{x+5}{(x-1)(x^2-1)}$ are of the form (a) $\frac{A}{x-1} + \frac{B}{x^2+1}$ (b) $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$ (c) $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$ (d) NOTA
- Partial fractions of $\frac{1}{x^2-1}$ are equivalent to (a) $Ax + \frac{B}{x^2-1}$ (b) $\frac{A}{x+1} + \frac{B}{x-1}$ (c) $\frac{A}{x+1}$ (d) $\frac{B}{x-1}$

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32. Express $\frac{4\pi}{6}$ radian in degree. (a) 60° (b) 90° (c) 180° (d) 120°
33. Find the area of the sector of a circle radius 8cm having an angle $\frac{\pi}{3}$ rad. (a) $\frac{32\pi}{3}$ rad (b) $\frac{32\pi}{4}$ rad (c) $\frac{24\pi}{3}$ rad (d) $\frac{24\pi}{4}$ rad
34. Evaluate $\sin 75^\circ$ without the use of calculator. (a) $\frac{\sqrt{3}+2}{2\sqrt{2}}$ (b) $\frac{\sqrt{2}+3}{4}$ (c) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ (d) $\frac{1}{\sqrt{2}}$
35. If $\sec^2 \theta + \tan^2 \theta = 3$, the angles θ is (a) 45° (b) 60° (c) 105° (d) 30°
36. Evaluate $\tan 105^\circ$ in surd form. (a) $2 - \sqrt{3}$ (b) $\sqrt{3} - 2$ (c) $2 + \sqrt{3}$ (d) $-2 - \sqrt{3}$
37. If $\tan 2\theta = 5$ find θ . (a) 14° (b) 15° (c) 18° (d) 180°
38. If $\sin^2 \alpha + \cos^2 \alpha = 1$, find $\cos^2 \alpha + \sin^2 \alpha$. (a) 2 (b) $2 - 2\sin^2 \alpha$ (c) $1 - \cos^2 \alpha$ (d) $1 - \sin^2 \alpha$
39. If a is the adjacent of right hand triangle, b is the opposite and c is the hypotenuse, then what $\sin \alpha$, if α is the angle connecting hypotenuse and adjacent sides. (a) $\frac{b}{c}$ (b) $\frac{a}{c}$ (c) $\frac{a}{b}$ (d) $\frac{c}{a}$
40. If $\cos^2 \theta = \frac{25}{9}$ then, what is $\cos \theta$. (a) $\frac{10}{3}$ (b) 6 (c) 8 (d) $\frac{5}{3}$
41. If $\cos^2 \theta = \frac{36}{4}$ find $\sin \theta$. (a) 3 (b) $\frac{6}{4}$ (c) 13 (d) 2
42. If $\sin^2 \theta + \cos^2 \theta = 1$, find $1 - \sin^2 \theta$. (a) $\cos^2 \theta$ (b) $\sin^2 \theta$ (c) 30° (d) 45°
43. Express $\sin 2A$ in multiple angles formulae. (a) $2\sin A \cos A$ (b) $\cos A + \sin A$ (c) $\sin A$ (d) $\cos A \sin A$
44. The set of natural numbers include all except (a) $\{1, 2, 3\}$ (b) $\{1, 2, 3, \dots\}$ (c) $\{x \in \mathbb{N} : 0 \leq x \leq \infty\}$ (d) $\{x \in \mathbb{N} : x \geq 1\}$
45. The nonnegative integer is equivalent to (a) $\{x : x \geq 0\}$ (b) $\{x \in \mathbb{Z} : 0 < x < 1\}$ (c) 0 (d) NOTA
46. The set of real numbers is defined as (a) the x-axis (b) The number line (c) $\mathbb{N} \cup \mathbb{Z} \cup \mathbb{Q}$ (d) all of the above
47. If $\mathbb{Z}$ represents the set of integers, then $\{y : -3 < y < 2\}$ is (a) $\{-3, 2\}$ (b) $\{0, 1\}$ (c) $\{-3, -2, -1, 0, 1, 2\}$ (d) $\{-2, -1, 0, 1\}$
48. The n th term of the series $2+6+12+20+30+\dots$ is (a) $n^2 - n$ (b) $\frac{1}{2}n + 1$ (c) $2n + \frac{n^2}{2}$ (d) $n(n+1)$
49. Solve the following absolute value equation $|2x+1| = 7$. (a) $\{-2\}$ (b) $\{-9\}$ (c) $\{-2, -9\}$ (d) No solutions
50. The expression $1 + \frac{1}{4} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$ is NOT true for all $n \in \mathbb{Z}^+$ at (a) $n=2$ (b) $n=1$ (c) $n=3$ (d) NOTA
51. Compute $\frac{1}{n} \sum_{k=1}^n k$. (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(2n+1)}{2}$ (c) $\frac{n(n+1)}{3}$ (d) $\frac{n+1}{2}$
52. Find the sum of $\sum_{T=1}^T (T^2 - 2T + 2)$. (a) 10 (b) 12 (c) 18 (d) 20
53. If $T_n = (1 + 2n) - 5n^2$ find $|T_2 - T_3|$. (a) 76 (b) -76 (c) -152 (d) 152
54. Given $T_n = 5 - (4 + 2n^2)$ compute $|\sum_{n=1}^3 T_n|$. (a) -25 (b) -17 (c) 25 (d) 17
55. The 9th term of an AP is 3 times the 5th term. Find d if the 5th term is 4. (a) 1 (b) 2 (c) 0 (d) 3
56. How many terms of the sequence 6, 8, 10, ... will add up to 50? (a) 8 (b) 7 (c) 6 (d) 5
57. If $a=1/2, r=1/2$, find the sum of the GP at infinity. (a) $1/2$ (b) 1 (c) ∞ (d) 0
58. Find $7(3x^2 + 2x - 5) + 2(-x^2 + 3)$. (a) $20x^2 + 14x - 32$ (b) $19x^2 + 2x - 2$ (c) $19x^2 + 14x - 19$ (d) NOT
59. Solve for the unknowns $9x^2 + 12x - 3 = 0$. (a) $\{-3, \frac{5}{3}\}$ (b) $\{1, -1\}$ (c) $\{\frac{2}{3} \pm \pm 6\sqrt{7}\}$ (d) $\frac{-2 \pm \sqrt{7}}{2}$
60. Solve $2x + 1 \leq 3x - 2$, write the solution set in interval. (a) $(-\infty, -3]$ (b) $[3, \infty)$ (c) $(-\infty, 3]$ (d) $[-3, \infty)$
61. Solve $x^2 - 2x - 35 < 0$ and write the solution set in interval. (a) $(-5, 7)$ (b) $(0, 7)$ (c) $[-5, 7]$ (d) $(5, -7)$
62. Solve $5x^2 - 6x + 5 = 0$. (a) $\{0.6 \pm i0.8\}$ (b) $\{0.6 \pm i0.6\}$ (c) $\{0.8 \pm i0.6\}$ (d) $\{0.8 \pm i0.8\}$
63. Given $ax^2 + by + c = 0$ if $b^2 - 4ac < 0$ then the roots are? (a) distinct (b) equal (c) complex (d) undefined
64. If α and β are the roots of the equation $3x^2 + 5x - 2 = 0$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) $\frac{5}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $\frac{7}{2}$
65. If $A = \{x : x \leq 3\}$ and $B = \{x : x \geq 4\}$ find $A - B$ in interval form. (a) $-\infty < x \leq 3$ (b) $-\infty < x < 3$ (c) $-\infty \leq x \leq 3$ (d) $-\infty \leq x < 3$
66. Find the cardinality of the set of natural numbers. (a) 9 (b) 1 (c) ∞ (d) empty set
67. Which of the following statement is NOT true of the empty set? (a) The empty set is the collection of disjoint set (b) the empty set is an improper subset of a set (c) the empty set has cardinality zero (d) none of the above
68. If $A = \{x : x < 3\}$ and $B = \{x : x \geq 10\}$ find $A \cap B$. (a) $\{10\}$ (b) $\{0\}$ (c) $\{\}$ (d) NOTA
69. Evaluate $2\sqrt{8} - 3\sqrt{32} + 4\sqrt{50}$. (a) $12\sqrt{2}$ (b) $3\sqrt{90}$ (c) $6\sqrt{40}$ (d) $\sqrt{90}$
70. The first term of AP is 4 and its last term is -73. Find the number of terms if the sum of the sequence is 924. (a) 21 (b) 21 (c) 73 (d) 60



FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2018/2019
Course code & title: MTH 101 - General Mathematics I



NAME: _____
DEPARTMENT: _____

*TYPE H

MATRIC NO: _____
LEVEL: _____

Instruction: Answer all questions in this sheet using biro. Provide the correct answer to each question in the space provided.

- The n th term of the series $\sum_{n=1}^{\infty} n^3$ is
 - What is the coefficient of the 3rd term in the expansion of $(2x + \frac{3y}{4})^3$?
 - Find the coefficient of y^4 in the binomial expansion of $(1 - 2y)^6$
 - Absolute value of the following integers -5, -1, 0, 2, 4 in descending order can be best expressed as
 - The value of n when $5^{2n} + 3n - 1$ is an integer multiple is
 - The coefficient of x^3 in the expansion of $(1 + x)^{\frac{3}{2}}$ is
 - Given the sequence $-\frac{1}{2}, \frac{3}{2}, \frac{7}{2}, \frac{11}{2}, \dots$. Find its n th term.
 - The third term of an arithmetic series is -5 and the fifth term is 4.5. Find the 13th term.
 - If 19, x , y , -8 is an AP. What is (x, y) ?
 - The sum of the first n terms of an AP is 252. If the first term is -16 and the last term is 72, find the number of terms n
 - The n th term of the sequence 2, 6, 12, 20, 30, ... is
 - Find the roots of the equation: $x^3 - 2x^2 + x + 2 = 0$
 - For what value of k does the equation $3x^2 - 6x + k = 0$ have equal roots
 - If α and β are the roots of the equation $3x^2 + 5x - 2 = 0$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$
 - Evaluate $\tan 22.5^\circ$ without the use of a calculator.
 - Write $\cos 5x + \cos x$ as a product of two trig. Functions
- Use the following to answer 17 and 18. Given that $\cos \theta = \frac{4}{5}$
- Find $\sin \theta$
 - Find $\tan \theta$
 - $1 + \tan^2 \theta$ is equivalent to
 - Find the area of a triangle with sides 10cm, 7cm and 5cm.
 - Evaluate $\tan 75^\circ$ without the use of a calculator.
 - If $\tan 2\theta = 5$, find $\tan \theta$
 - Express $\frac{4\pi}{6}$ radian in degrees.
 - Find the area of the sector of a circle radius 8cm having an angle $\frac{\pi}{3}$
 - Evaluate $\frac{\sqrt{3} \times \sqrt{8} \times \sqrt{39}}{\sqrt{24} \times \sqrt{26}}$
- Use the following to answer 26, 27 and 28. Given that $\sin \alpha = \frac{5}{13}$
- Evaluate $\cos^2 \alpha$
 - Evaluate $\cos \alpha + \tan \alpha$
 - Evaluate $\operatorname{cosec}^2 \alpha$
 - If set $A = \{2, 9, 5, 2\}$, then the number of subsets which can be formed from A is
 - The cardinality of $B = \{5, 6, 3, 2, 3, 2\}$ is
 - If Z represents the set of integers, then $\{y: -3 < y < 2\}$ is
 - $A \subset B$ is read
 - If given the sets $A = \{1, 2, 3\}$ and $B = \{2, 1, 3\}$, what is the relationship between them?
 - The set difference of the set A with null set is
 - The relation $(A \cup B)'$ is equal to
 - Given the universal set $U = \{1, 2, 3, \dots, 10\}$, $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 5, 7, 9, 10\}$. Find $A \cup (B \cap C)$

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37. Find the number of subsets in the set $P = \{1, a, b, 3, 5\}$
38. If Q is a non-empty set and $\emptyset$ is an empty set, then $Q \cap \emptyset$ is Find $(P \cup Q)'$?
39. If $U = \{x: x \text{ is a natural number and } 1 \leq x \leq 9\}$, $P = \{x: 1 \leq x \leq 4\}$ and $Q = \{2, 4, 6, 8\}$. Find $(P \cup Q)'$?
40. P and Q are non empty sets of a universal set U where $U = \{\text{integers}\}$, $P = \{x: 2x + 6 > 0\}$ and $Q = \{x: (x+5)(x-2) < 0\}$. Find $P \cap Q$
41. Let A and B be sets of real numbers. If $n(A) = 23$ and $n(B) = 12$, $n(A \cup B) = 35$, find $n(A \cap B)$
42. $\frac{5}{6}$ of a class of 60 students study chemistry, $\frac{3}{5}$ study biology. Every student studies at least one of the subjects. How many students study both subjects?
43. Find the coefficient of x^3 in $(x - \frac{1}{3x^2})^9$
44. Solve for the middle term in the expansion $(\frac{a}{2} - \frac{b}{3})^{10}$
45. Write the first four terms of the expansion $(1 - 2x)^{-1}$
46. What is the fifth term in the expansion of $(x^3 - \frac{1}{2x})^5$?.....
47. Expand up to four terms $(4 + x)^{\frac{1}{2}}$
48. Expand $(2 + \frac{x}{3})^4$
49. Find the term independent of x in the expansion $(2x^2 + \frac{1}{x})^9$
- Use the following to answer 50 and 51. Given the fraction $\frac{8x^2-12}{x(x^2+2x-6)}$
50. Write the form of the partial fraction.....
51. Determine the partial fraction of the fraction.....
52. Resolve $\frac{5x+4}{(x+1)(x-2)}$ into partial fraction.....
53. Find the partial fraction decomposition of the expression $\frac{2x+3}{x^2+x}$
- Use the following to answer 54 and 55. Given the fraction $\frac{x^2+1}{x^3(x^2-1)^2}$
54. Write the form of the partial decomposition.....
55. Determine the partial fraction.....
56. Resolve $\frac{3x^2+4x+5}{(x+1)(x-3)(2x-1)}$ into partial fraction.....
- Use the following to answer 57, 58 and 59. Given the complex number $z = 4 - 3i$
57. Find the modulus.....
58. Find the argument.....
59. Express the complex number in its polar form.....
60. Evaluate $1 + i + i^2 + i^3 + \dots + i^{41}$ and simplify in the standard form $a + bi$
61. Simplify the complex number $\frac{1}{\cos \theta + i \sin \theta}$
62. Evaluate $\frac{10+5i}{3-4i}$
63. Simplify the complex number $(x + yi)^3 + (x - yi)^3$
64. Express $z = 3(\cos 30^\circ + i \sin 30^\circ)$ in exponential form.....
65. If $z = 3 + i\sqrt{3}$. Compute z^2 in its polar form.....
66. Express $e^{1-i\frac{\pi}{6}}$ in the form $a + ib$
67. Evaluate $(\cos 20^\circ + i \sin 20^\circ)(\cos 40^\circ + i \sin 40^\circ)$
68. Find the quadratic equation with the roots $1 + 2i$ and $1 - 2i$
69. Express $\frac{3+i}{2-i}$ in the form $x + iy$ where x and y are real.....
70. Compute $(5 + 2i)(3 - 3i)$

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DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018
Course code & title: MTH 111 (General Mathematics I)

Duration: 2 ½ hrs

TYPE H

Instruction: Answer all questions. Submit your question paper and your answer booklet at the end of the examination.

- Without using tables, evaluate $\frac{1+\tan 60^\circ \tan 30^\circ}{\tan 60^\circ \tan 30^\circ}$ (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{2}}{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{2\sqrt{2}}{3}$
- If $\sin \theta = \frac{m^2 - n^2}{m^2 + n^2}$, find $\cot \theta$. (a) $\frac{2mn}{m^2 - n^2}$ (b) $\frac{2mn}{m^2 + n^2}$ (c) $\frac{m^2 - n^2}{2mn}$ (d) $\frac{m^2 + n^2}{2mn}$
- Rationalize $\frac{1+\tan 60^\circ}{1-\tan 60^\circ}$ and express in surd form. (a) $2 + \sqrt{2}$ (b) $2 - \sqrt{2}$ (c) $-2 + \sqrt{2}$ (d) $-2 - \sqrt{2}$
- Given that $\sin \theta = \frac{3}{5}$, $\cos \alpha = \frac{4}{5}$, find $\sin(\theta + \alpha)$. (a) 1 (b) $\frac{24}{25}$ (c) $\frac{9}{25}$ (d) $\frac{16}{25}$
- If $\cos x = \sqrt{\frac{a}{b}}$, find $\operatorname{cosec} x$. (a) $\sqrt{\frac{b}{b-a}}$ (b) $\sqrt{\frac{b}{a-b}}$ (c) $\sqrt{\frac{a}{b-a}}$ (d) $\sqrt{\frac{a}{a-b}}$
- Find the values of x and y between 0° & 180° which satisfy the simultaneous equations $\sin(x+y) = \frac{\sqrt{2}}{2}$ & $\cos 2x = -\frac{1}{2}$ (a) $x = -15^\circ, y = 60^\circ$ (b) $x = 15^\circ, y = 60^\circ$ (c) $x = 60^\circ, y = -15^\circ$ (d) $x = -60^\circ, y = 15^\circ$
- Use the following to answer questions 7 and 8. Using the half-angle substitution $(\tan \frac{1}{2}\theta = t)$.
- Find $\sec \theta \cot \theta = 1$ in terms of t . (a) $\frac{(1-t^2)}{(1+t^2)} \cdot \frac{(2t)}{(1-t^2)} = 1$ (b) $\frac{(1-t^2)}{(1+t^2)} \cdot \frac{(1+t^2)}{(2t)} = 1$ (c) $\frac{(1+t^2)}{(1-t^2)} \cdot \frac{(2t)}{(1+t^2)} = 1$ (d) $\frac{(1+t^2)}{(1-t^2)} \cdot \frac{(1-t^2)}{(2t)} = 1$
- Find the value(s) of t from your answer in (12). (a) ± 1 (b) 1 twice (c) -1 twice (d) $\frac{\sqrt{3}}{3}$
- Solve for t in $\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1$ (a) $t = 0$ or 1 (b) $t = 0$ or -1 (c) $t = \pm 1$ (d) -1 twice
- Solve for θ in the equation $3 \tan \theta = 2 \cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. (a) $30^\circ, 150^\circ$ (b) $60^\circ, 150^\circ$ (c) $30^\circ, 120^\circ$ (d) $60^\circ, 120^\circ$
- If Kx^4 is a term in the binomial expansion of $(2+3x)^5$. Find the value of K . (a) 810 (b) 720 (c) 1080 (d) 240
- Rationalize $\frac{\sqrt{6}-\sqrt{4}}{\sqrt{6}+\sqrt{4}}$. (a) $5 + 2\sqrt{6}$ (b) $5 - 4\sqrt{6}$ (c) $5 + 4\sqrt{6}$ (d) $5 - 2\sqrt{6}$
- Find the sum of the series $\sum_{n=1}^{\infty} (2n-1)^2$ (a) $\frac{n(4n^2-1)}{3}$ (b) $\frac{n(n+1)^2}{2}$ (c) $\frac{1}{2}n(n+1)$ (d) $\frac{1}{2}n^2(n-1)$
- Determine the middle term of expansion $(a-2b)^8$. (a) $-1792a^3b^5$ (b) $1120a^4b^4$ (c) $-448a^5b^3$ (d) $1872a^2b^2$
- Find the 3rd term in the expansion $(1-2x)^{-2}$. (a) $12x^2$ (b) $\frac{1}{16}x^3$ (c) $\frac{1}{24}x^3$ (d) $\frac{1}{12}x^3$
- Given the series $1 + 4 + 9 + \dots + n^2$. The sum is defined by. (a) $\frac{n(n+1)^2}{2}$ (b) $\frac{n(n+1)(2n+1)}{6}$ (c) $\frac{1}{2}n(n+1)$ (d) $\frac{1}{2}n^2(n-1)$
- The ratio of the coefficient of x^4 to x^3 in the binomial expansion $(1+2x)^m$ is 3:1. Find the value of m . (a) 1 (b) 3 (c) 7 (d) 9
- Evaluate $(3 + \sqrt{5})^3 - (3 - \sqrt{5})^3$. (a) 0 (b) 72 (c) $64\sqrt{5}$ (d) $32\sqrt{5}$
- Find the sum of n positive integers. (a) $\left(\frac{n(n+1)}{2}\right)^2$ (b) $\frac{n(2n+1)(n+1)}{2}$ (c) $\frac{1}{2}n^2(n+1)$ (d) $\frac{1}{2}n^2(4n-1)$
- Find m , given $5\sqrt{3} - \sqrt{147} + 2\sqrt{m} = 0$. (a) 1 (b) 3 (c) 5 (d) 7
- Express the roots of $x^2 + 4x + 40 = 0$ in the form $a + bi$. (a) $2+6i, 2-6i$ (b) $-2+6i, -2-6i$ (c) $2+12i, 2-12i$ (d) $-2+12i, -2-12i$
- If $\tan(A-B) = \frac{1}{5}$ and $\tan B = 2$, find $\tan A$. (a) $11/5$ (b) $11/3$ (c) $3/11$ (d) $5/11$
- Use the following to answer questions 23, 24, and 25. Given the complex number $5\sqrt{3} - 5i$.
- Find the modulus (a) 5 (b) $\sqrt{10}$ (c) 10 (d) $\sqrt{7}$
- Find the argument (a) 300° (b) 180° (c) -30° (d) 150°
- Find the polar form (a) $5(\cos 180^\circ + i \sin 180^\circ)$ (b) $10(\cos 150^\circ + i \sin 150^\circ)$ (c) $\sqrt{10}(\cos 300^\circ + i \sin 300^\circ)$ (d) $\sqrt{7}(\cos 30^\circ - i \sin 30^\circ)$
- Evaluate the complex number $\frac{-1+3i}{2+i}$. (a) $\frac{1}{5} + \frac{7}{5}i$ (b) $\frac{-1}{5} + \frac{4}{5}i$ (c) $\frac{3}{5} + \frac{2}{5}i$ (d) $\frac{7}{5} + \frac{1}{5}i$
- Express $\frac{4\pi}{6}$ radian in degrees. (a) 60° (b) 90° (c) 150° (d) 120°
- In a school of 150 students, 80 offer French while 60 offer Arabic and 20 offers neither. How many students offer both subjects? (a) 45 (b) 10 (c) 35 (d) 30
- Given $U = \{x: x \text{ is a positive integer less than } 15\}$ and $P = \{x: x \text{ is an even number from } 1 \text{ to } 14\}$. Find the complement of P . (a) $\{1, 3, 5, 7, 9, 11, 13, 15\}$ (b) $\{2, 3, 5, 7, 9, 11, 13\}$ (c) $\{1, 3, 5, 7, 9, 11, 13\}$ (d) $\{2, 3, 5, 7, 11, 15\}$
- The symmetric difference between A & B if $A = \{1, 2, 3, 4\}$ & $B = \{2, 4, 6, 8\}$ is (a) $\{2, 6\}$ (b) $\{6, 8\}$ (c) $\{1, 3, 6, 8\}$ (d) $\{2, 4, 8\}$
- One root of the equation $27x^2 + bx + 8 = 0$ is known to be the square of the other. Find b . (a) 35 (b) 19 (c) 216 (d) -30
- Construct an equation with roots $\sqrt{2} + 1, \sqrt{2} - 1$. (a) $x^2 - 2\sqrt{2}x + 1 = 0$ (b) $\sqrt{2}x^2 - 2\sqrt{2}x + 2\sqrt{2} = 0$ (c) $2\sqrt{2}x^2 - 2\sqrt{2}x + 1 = 0$ (d) $2x^2 - 2\sqrt{2}x + 2 = 0$
- Solve the equation $y - 11\sqrt{7} + 24 = 0$ (a) $y = 8$ or 3 (b) 24 or 9 (c) $y = 64$ or 9 (d) $y = 12$ or 4
- Find the values of m which makes the following quadratic function a perfect square. $x^2 + 2(m+1)x + m + 3$. (a) $m = -2$ or (b) $m = -1$ or -3 (c) $m = 1$ or -1 (d) $m = 1$ or 3
- The solution of the quadratic equation $6x^2 - 10x + 4 = 0$ is (a) $x = 3$ or 5 (b) $x = 1$ or 3 (c) $x = 1$ or $\frac{2}{3}$ (d) $x = \frac{1}{3}$ or $\frac{2}{3}$

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37. If α and β are the roots of the equation: $2x^2 - 6x + 5 = 0$. Evaluate $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) $\frac{5}{2}$ (b) $\frac{3}{2}$ (c) $\frac{5}{4}$ (d) $\frac{3}{4}$
38. Given the sequence 5, 1, -3, ..., which term of the sequence is -200? (a) 17th term (b) 16th term (c) 28th term (d) none of the above
39. Answer questions 39 - 41 using the following. Given the fraction $\frac{3x-5}{(x+1)^2(x-3)^2}$.
The partial fraction will be of the form? (a) $\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3} + \frac{D}{(x-3)^2}$ (b) $\frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-3} + \frac{D}{(x-3)^2}$ (c) $\frac{Ax+B}{(x+1)^2} + \frac{Cx+D}{(x-3)^2}$ (d) $\frac{Ax+B}{(x+1)^2} + \frac{Cx+D}{(x-3)^2}$
40. Find (A, B). (a) $(-\frac{5}{32}, -\frac{1}{2})$ (b) $(-\frac{1}{2}, -\frac{5}{32})$ (c) $(\frac{1}{4}, -\frac{7}{32})$ (d) $(\frac{1}{2}, \frac{1}{4})$
41. Find (C, D). (a) $(\frac{1}{2}, -\frac{1}{4})$ (b) $(-\frac{7}{32}, \frac{1}{4})$ (c) $(\frac{2}{3}, -1)$ (d) $(-\frac{1}{2}, -\frac{5}{32})$
42. Given the sequence 5, 1, -3, ..., which term of the sequence is -75? (a) 7th term (b) 14th term (c) 21st term (d) 30th term
43. The 6th and 13th term of an A.P. are 0 and 14 respectively. Find the A.P. (a) -10, -8, -6, ... (b) 6, 8, 10, ... (c) 0, 14, 28, ... (d) 14, 28, 32, ...
44. The sum of three consecutive terms of an A.P. is 18 and their product is 180, the common difference is a positive integer. Find the terms. (a) 10, 6, 2, ... (b) 2, 6, 10, ... (c) 18, 28, 38, ... (d) 12, 18, 25, ...
45. What is the 15th term of the sequence -3, 2, 7, ...? (a) 67 (b) 42 (c) 28 (d) 72
46. Find the 12th term of the sequence 128, 64, 32, ... (a) 150 (b) 144 (c) $\frac{1}{16}$ (d) $\frac{1}{2}$
47. The 4th and 8th terms of a G.P. are 24 and $\frac{8}{27}$ respectively. Find the first term if the common ratio is a positive real number. (a) 72 (b) 648 (c) 72 (d) 216
48. The sum of 8 terms of an A.P. is 12 and the sum of 16 terms is 56. Find the A.P. (a) $\frac{2}{9}, \frac{3}{7}, \frac{8}{12}, \dots$ (b) 12, 36, 72, ... (c) 12, 24, 36, ... (d) $-\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \dots$
49. Find the sum of the first ten terms of $\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, \dots$ (a) $127\frac{7}{8}$ (b) 124 (c) $\frac{1}{64}$ (d) $\frac{1}{640}$
50. Find the sum to infinity of $20 + 12 + \frac{36}{5} + \dots$ (a) 42 (b) 50 (c) 32 (d) 60
51. If the equation $x^2 - 9x + 1 = p(x-3)$ has equal roots, find the possible values of p. (a) 3 or $-\frac{1}{3}$ (b) 3 or -1 (c) 5 or -1 (d) none of these
52. Find the value of $\cot^2 \theta$ if θ is acute and $\cos \theta = \frac{1}{2}$. (a) 3 (b) $\frac{1}{9}$ (c) $\sqrt{3}$ (d) $\frac{1}{3}$
- Use the following to answer questions 53, 54 and 55. If $\sin \alpha = \frac{5}{13}$ in a right-angle triangle, evaluate
53. $\operatorname{cosec}^2 \alpha$. (a) $\frac{25}{144}$ (b) $\frac{169}{25}$ (c) $\frac{156}{209}$ (d) $\frac{169}{144}$
54. $\cot^2 \alpha$. (a) $\frac{25}{144}$ (b) $\frac{169}{25}$ (c) $\frac{156}{209}$ (d) $\frac{144}{25}$
55. $\sec^2 \alpha + \sin^2 \alpha$. (a) $\frac{25}{144} + \frac{25}{169}$ (b) $\frac{169}{144} + \frac{169}{25}$ (c) $\frac{25}{169} + \frac{169}{144}$ (d) $\frac{169}{144} + \frac{205}{156}$
56. Find the values of x for which $2 \cos x - 1 = 0$; $0^\circ < x < 360^\circ$ (a) 140° or 180° (b) 120° or 180° (c) 120° or 240° (d) 140° or 210°
57. The set $(A' \cap B')$ is equivalent to (a) $B - A$ (b) $A' \cap B$ (c) $B' \cap A'$ (d) $A - B$
- Use the following to answer questions 58, 59, 60 & 61. If $z_1 = 2 + i$, $z_2 = 3 - 2i$ and $z_3 = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$, evaluate
58. $|3z_1 - 4z_2|$ (a) $\sqrt{166}$ (b) $\sqrt{157}$ (c) $\sqrt{127}$ (d) $\sqrt{176}$
59. $z^3 - 3z^2 + 4z - 8$ (a) $2 + 5i$ (b) $-4 + 9i$ (c) $-7 + 3i$ (d) $6 - 5i$
60. $(z_3)^4$ (a) $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ (b) $-\frac{1}{2} + \frac{\sqrt{3}}{2}i$ (c) $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$ (d) $-\frac{1}{2} - \frac{\sqrt{3}}{2}i$
61. $\frac{2z_2 + z_1 - 5 - i}{2z_1 - z_2 + 3 - i}$ (a) 1 (b) 3 (c) 7 (d) $3 + 5i$
62. Solve the equation: $x + 2\sqrt{x} - 8 = 0$. (a) (1, 2) (b) (2, -3) (c) (4, 16) (d) (2, -4)
63. Given $x + y = 3$ and $y^2 + x^2 - 2y - 1 = 0$. Find (x, y). (a) (1, 2) (b) (1, -2) (c) (2, 3) (d) $(\frac{1}{2}, -1)$
64. Resolve $\frac{3x}{(1+x)^3}$ into partial fraction (a) $\frac{1}{(1+x)^2} - \frac{1}{(1+x)^3}$ (b) $\frac{2}{(1+x)^2} - \frac{2}{(1+x)^3}$ (c) $\frac{3}{(1+x)^2} - \frac{3}{(1+x)^3}$ (d) $\frac{1}{(1+x)^2} - \frac{1}{(1+x)^3}$
65. The sum of the first three terms of a G.P. is half its sum to infinity. Find the positive common ratio of the progression? (a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{5}$
66. Simplify $\frac{3(2^{n+1}) - 4(2^{n-1})}{2^{n+1} - 2^n}$ (a) 2^{n+1} (b) 2^{n-1} (c) 4 (d) $\frac{1}{4}$
67. Find the roots of the equation $x^3 - 2x^2 - x + 2 = 0$. (a) (2, 1, -3) (b) (2, 1, 3) (c) (-1, 2, 3) (d) (-1, 1, 2)
68. Resolve $\frac{x-1}{(x+2)(x+3)}$ into partial fractions. (a) $\frac{2}{(x+2)} - \frac{1}{(x+3)}$ (b) $\frac{3}{(x+2)} + \frac{2}{(x+3)}$ (c) $\frac{3}{5(x+2)} + \frac{2}{5(x+3)}$ (d) $\frac{5}{(x+2)} - \frac{7}{(x+3)}$
69. Solve for the coefficient of x^2y^3 in the expansion of $(2x + 3y)^5$. (a) 540 (b) 2160 (c) 1080 (d) 270
70. What is the coefficient of r^3 in the series expansion of $(1+r)^{\frac{1}{2}}$? (a) $\frac{1}{16}$ (b) $\frac{2}{27}$ (c) $-\frac{1}{27}$ (d) $\frac{5}{81}$

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COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS / COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2016/2017
 Course code & title: MTH 111 (General Mathematics I) Duration: 2 ½ hrs
TYPE D

Instruction: Answer all questions. Submit your question paper and your answer booklet at the end of Examination.

1. Evaluate $\tan 22.5^\circ$ without the use of a calculator. (a) $\sqrt{2} - 1$ (b) $1 - \sqrt{2}$ (c) $\sqrt{2} - 2$ (d) $2 - \sqrt{2}$
2. Write $\cos 3x + \cos x$ as a product of two trig. Functions (a) $\cos 3x \cos 2x$ (b) $2 \cos 3x \sin 2x$ (c) $2 \cos 3x \cos 2x$ (d) $\sin 3x \cos 2x$
3. Express $\frac{10+15i}{5-4i}$ in the form $a + bi$ (a) $\frac{-6+17i}{5}$ (b) $\frac{18+17i}{5}$ (c) $\frac{17+6i}{5}$ (d) $\frac{18+6i}{5}$
4. Find the value of i^{54} in complex numbers (a) $\sqrt{-1}$ (b) 1 (c) -1 (d) $\sqrt{-i}$
5. If $A = \{(x, y): y^2 = 4x\}$ and $B = \{(x, y): y^2 = 16\}$, find the elements of $A \cap B$ (a) $\{4, 16\}$ (b) $\{2, 4\}$ (c) $\{8, 4\}$ (d) $\{2, 8\}$
6. Given that $U = \{x: -10 \leq x \leq 10\}$, $P = \{x: -10 < x < 10\}$ and $Q = \{x: -5 < x \leq 3\}$, which of the following is correct. I. $P' \cap Q = \emptyset$ II. $P \cup Q = U$ III. $P' \subset Q'$ (a) I and II (b) II and III (c) I and III (d) I, II, and III
7. $|3x + 5| = 8$. Find all values of x which make this equation true (a) $\{1, -\frac{4}{3}\}$ (b) $\{4\frac{1}{3}, -1\}$ (c) $\{1, -1\}$ (d) $\{4\frac{1}{3}, -4\frac{1}{3}\}$
8. The symmetric difference between sets A and B is (a) $(A-B) \cup (B-A)$ (b) $(A-B) \cap (B-A)$ (c) $(A-B) \cup B'$ (d) $A-B$
9. Use the following to answer 9 and 10. Given that $\cos \theta = \frac{1}{3}$
10. Find $\sin \theta$. (a) $2\sqrt{2}$ (b) $\frac{2\sqrt{2}}{3}$ (c) $\frac{3\sqrt{2}}{2}$ (d) $\frac{\sqrt{2}}{2}$
11. Find $\tan \theta$. (a) $2\sqrt{2}$ (b) $\frac{2\sqrt{2}}{3}$ (c) $\frac{3\sqrt{2}}{2}$ (d) $\frac{\sqrt{2}}{2}$
12. $1 + \tan^2 \theta$ is equivalent to (a) $1 + \cot^2 \theta$ (b) $\operatorname{cosec}^2 \theta$ (c) $\sec^2 \theta$ (d) $1 + \sec^2 \theta$
13. Multiply $3 + 4i$ and $5 + 7i$ in complex numbers and simplify. (a) $41 + 13i$ (b) $-41 + 13i$ (c) $13 + 41i$ (d) $-13 + 41i$
14. Use the following to answer 13, 14, and 15. Given the complex number $2 + 3i$
15. Find the modulus (a) $\sqrt{5}$ (b) $\sqrt{10}$ (c) $\sqrt{13}$ (d) $\sqrt{7}$
16. Find the argument (a) 53.6° (b) 56.3° (c) 63.6° (d) 66.3°
17. Find the polar form (a) $\sqrt{5}(\cos 53.6^\circ + i \sin 53.6^\circ)$ (b) $\sqrt{13}(\cos 56.3^\circ + i \sin 56.3^\circ)$ (c) $\sqrt{10}(\cos 63.6^\circ + i \sin 63.6^\circ)$ (d) $\sqrt{7}(\cos 66.3^\circ + i \sin 66.3^\circ)$
18. Express $\frac{4\pi}{6}$ radian in degrees. (a) 60° (b) 90° (c) 150° (d) 120°
19. Find the area of the sector of a circle radius 8cm having an angle $\frac{\pi}{3}$ (a) $\frac{32\pi}{3}$ rad (b) $\frac{32\pi}{4}$ rad (c) $\frac{24\pi}{3}$ rad (d) $\frac{24\pi}{4}$ rad
20. Evaluate $\frac{\sqrt{3} \times \sqrt{8} \times \sqrt{20}}{\sqrt{24} \times \sqrt{26}}$ (a) $\frac{3}{\sqrt{6}}$ (b) $\frac{\sqrt{3}}{6}$ (c) $\frac{\sqrt{5}}{2}$ (d) $\frac{2}{\sqrt{6}}$
21. Evaluate $2\sqrt{3} - 3\sqrt{32} + 4\sqrt{50}$ (a) $12\sqrt{2}$ (b) $3\sqrt{90}$ (c) $6\sqrt{40}$ (d) $\sqrt{90}$
22. Let $M = \{\text{all positive even integers less than or equal to } 20\}$, $N = \{\text{all factors of } 20\}$. What is $M \cap N$? (a) $\{1, 2, 4, 5, 10, 20\}$ (b) $\{2, 4, 10, 20\}$ (c) $\{6, 8, 10, 12, 14\}$ (d) $\{1, 4, 10, 20\}$
23. The set of difference of the set A with null set is (a) A (b) null set (c) universal set (d) none of the above
24. Which of the following is false? (a) $(A \cup B) \cup C = A \cup (B \cup C)$ (b) $(A \cup B)' = A' \cup B'$ (c) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (d) $A \cap (A \cup B) = A$
25. Given the universal set $U = \{1, 2, 3, \dots, 10\}$, $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 5, 7, 9, 10\}$. Find $A \cup (B \cap C)$ (a) $\{3, 5, 7\}$ (b) $\{3, 5, 7, 9, 10\}$ (c) $\{1, 2, 3, \dots, 10\}$ (d) $\{2, 4, 5, 6, 7, 8, 9, 10\}$
26. Partial fractions of $\frac{x^2+1}{x^2+1}$ will be of the form (a) $\frac{A}{x-1} + \frac{B}{x^2-x+1}$ (b) $\frac{A}{x+1} + \frac{B}{x^2-x+1}$ (c) $\frac{A}{x+1} + \frac{Bx+C}{x^2-x+1}$ (d) none of the above
27. Resolve $\frac{x}{(x+2)(x-3)}$ into partial fractions (a) $\frac{2}{5(x+2)} - \frac{3}{5(x-3)}$ (b) $\frac{2}{5(x+2)} + \frac{3}{5(x-3)}$ (c) $\frac{2}{5(x-2)} + \frac{3}{5(x-3)}$ (d) $\frac{3}{5(x-3)} - \frac{2}{5(x+2)}$
28. If $X = \{\text{all perfect squares less than } 40\}$ and $Y = \{\text{all odd numbers from } 1 \text{ to } 15\}$. Find $X \cap Y$. (a) $\{9\}$ (b) $\{1, 9\}$ (c) $\{3, 9\}$ (d) $\{9, 25\}$
29. $\frac{1}{(x-3)(x-2)}$ is equal to (a) $\frac{1}{x^2} - \frac{1}{x-2}$ (b) $\frac{1}{x-3} - \frac{1}{x-2}$ (c) $\frac{1}{x+3} - \frac{1}{x+2}$ (d) $\frac{1}{x^2} - \frac{1}{x-2}$
30. Partial fractions of $\frac{x+5}{(x-1)(x^2-1)}$ are of the form (a) $\frac{A}{x-1} + \frac{B}{x^2+1}$ (b) $\frac{A}{x-1} + \frac{Bx+C}{x^2+1}$ (c) $\frac{A}{x+1} + \frac{Bx+C}{x^2+1}$ (d) none of the above
31. Partial fractions of $\frac{1}{x^2-1}$ are equivalent to (a) $Ax + \frac{B}{x^2-1}$ (b) $\frac{A}{x+1} + \frac{B}{x-1}$ (c) $\frac{A}{x+1}$ (d) $\frac{B}{x-1}$
32. Given $P = \{1, 3, 5, 7, 9, 11\}$ and $Q = \{2, 4, 6, 8, 10, 12\}$. Determine the relationship between P and Q (a) $P \subset Q$ (b) $Q \subset P$ (c) $P = Q$ (d) $P \cap Q = \emptyset$
33. Partial fractions of $\frac{x}{(x-a)(x-b)(x-c)}$ will be of form (a) $\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$ (b) $\frac{A}{x+a} + \frac{B}{x+b} + \frac{C}{x+c}$ (c) $\frac{A}{x+a} + \frac{B}{x-b} + \frac{C}{x+c}$ (d) $\frac{A}{x-a} + \frac{Bx+C}{(x-b)(x-c)}$

32. Absolute value of the following integers $-5, -1, 0, 2, 4$ in descending order can be best expressed as (a) 5, 4, 2, 1, 0 (b) 0, 1, 2, 4, 5
(c) 2, 4, 0, -5, -1 (d) 0, -1, 2, -5, 4
33. The distance between -55 and 10 on a real number line is (a) 45 (b) -45 (c) -550 (d) 65
34. Let a, b , and c be real numbers. Which of the following is false (a) $a + (b + c) = (a + b) + c$ (b) $a(b + c) = ab + ac$ (c) $a(b - c) = ab - ac$
(d) $a(bc) = (ab)c$
35. Which of the following is not a subset of the set of real numbers (a) Natural numbers (b) Whole numbers (c) Irrational numbers
(d) complex numbers
36. Which of the following numbers is farthest from the number 1 in the number line (a) -11 (b) -5 (c) 5 (d) 11
37. Express 35° in radian measure (a) $\frac{36}{7\pi}$ rad (b) $\frac{35}{5\pi}$ rad (c) $\frac{7\pi}{36}$ rad (d) $\frac{5\pi}{35}$ rad
38. Find the area of a triangle with sides 10cm, 7cm and 5cm. (a) 17.25cm^2 (b) 16.25cm^2 (c) 18.25cm^2 (d) 15.25cm^2
39. Evaluate $\tan 75^\circ$ without the use of a calculator. (a) $2 + \sqrt{3}$ (b) $2 + \sqrt{2}$ (c) $3 + \sqrt{2}$ (d) $3 + \sqrt{3}$
40. If $\tan \theta = 5$, find $\tan \theta$. (a) $\frac{2+\sqrt{23}}{5}$ (b) $\frac{1+\sqrt{26}}{5}$ (c) $\frac{-2+\sqrt{26}}{5}$ (d) $\frac{-1+\sqrt{26}}{5}$
41. The quadratic equation whose roots are $2 - \sqrt{2}$ and $2 + \sqrt{2}$ is (a) $x^2 + x - 2 = 0$ (b) $5x^2 - 3x + 11 = 0$ (c) $2x^2 - 3x + 7 = 0$
(d) $x^2 - 4x + 2 = 0$
42. Solve the equations: $m^2 - n^2 = 9$, $m - n = 3$. (a) (1, 2) (b) (3, 0) (c) (-1, 3) (d) (2, 5)
43. Solve for y in the equation, $\sqrt{y+5} = 5 - \sqrt{y}$. (a) 1 (b) 2 (c) 3 (d) 4
44. For what value of k does the equation $3x^2 - 6x + k = 0$ have equal roots (a) 1 (b) 2 (c) 3 (d) 4
45. Find the value(s) of the number n such that twice its square is equal to three times the number plus 5. (a) $(-1, \frac{5}{2})$ (b) $(-2, \frac{1}{2})$ (c)
(-1, 2) (d) $(-2, \frac{2}{3})$
46. Given $ax^2 + by + c = 0$ if $b^2 - 4ac < 0$ then the roots are? (a) distinct (b) equal (c) complex (d) undefined
47. If α and β are the roots of the equation $3x^2 + 5x - 2 = 0$. Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. (a) $\frac{5}{2}$ (b) $\frac{3}{2}$ (c) $\frac{1}{2}$ (d) $\frac{7}{2}$
48. What is the coefficient of the 3rd term in the expansion of $(2x + \frac{3y}{4})^8$? (a) $\frac{27}{8}$ (b) $\frac{81}{8}$ (c) $\frac{27}{64}$ (d) $\frac{81}{64}$
49. Find the coefficient of y^4 in the binomial expansion of $(1 - 2y)^6$. (a) 280 (b) 240 (c) 200 (d) 180
50. Find the value of n if the ratio of the coefficient of y^4 to that of y^3 in the binomial expansion of $(1 + 2y)^n$ is 3:1. (a) 3 (b) 9 (c)
(d) 5
51. Using Pascal triangle evaluate $(1 + \sqrt{3})^4 - (1 - \sqrt{3})^4$. (a) $9\sqrt{3}$ (b) $27\sqrt{3}$ (c) $32\sqrt{3}$ (d) $16\sqrt{3}$
52. The n th term of the series $\sum_{i=1}^n n^3$ is (a) $\frac{n(n+1)(2n+1)}{6}$ (b) $\frac{n(4n^2-1)}{3}$ (c) $\left[\frac{n(n+1)}{2}\right]^2$ (d) $\frac{(2n-1)^2}{4}$
53. The value of n when $5^{2n} + 3n - 1$ is an integer multiple is (a) 2 (b) 3 (c) 5 (d) 7
54. The coefficient of x^3 in the expansion of $(1 + x)^{\frac{1}{2}}$ is (a) $\frac{1}{8}$ (b) $\frac{1}{16}$ (c) $\frac{1}{32}$ (d)
55. If $S_n = 2n(n + 3)$ is the sum of the first n terms of a series, find the 4th term. (a) 10 (b) 15 (c) 25 (d) 20
56. If set $A = \{2, 9, 5, 2\}$, then the number of subsets which can be formed from A is (a) 3 (b) 4 (c) 8 (d) 16
57. The cardinality of $B = \{5, 6, 3, 2, 3, 2\}$ is (a) 6 (b) 5 (c) 4 (d) 3
58. If Z represents the set of integers, then $\{x : -3 < x < 2\}$ is (a) $\{-3, 2\}$ (b) $\{0, 1\}$ (c) $\{-3, -2, -1, 0, 1, 2\}$ (d) $\{-2, -1, 0, 1\}$
59. In a survey of 50 newspapers, 40 read Champion and 30 read Guardian, how many read both papers? (a) 20 (b) 15 (c) 10 (d) 5
60. If $X = \{n^2 + 1 : n = 0, 2, 3\}$ and $Y = \{n + 1 : n = 2, 3, 5\}$ Find $X \cap Y$. (a) $\{1, 3\}$ (b) $\{5, 10\}$ (c) $\emptyset$ (d) $\{4, 6\}$
61. Given the sequence $-\frac{1}{2}, \frac{3}{2}, \frac{7}{2}, \frac{11}{2}, \dots$. Find its n th term. (a) $\frac{1}{2}(4n - 5)$ (b) $2n - 1$ (c) $5n - 2$ (d) $\frac{1}{2}(2n - 5)$
62. The third term of an arithmetic series is -5 and the fifth term is 4.5. Find the 13th term. (a) 33 (b) 42.5 (c) 14.5 (d) 47
63. If 19, x , y , -8 is an AP. What is (x, y) ? (a) (12, 7) (b) (7, 12) (c) (10, 1) (d) (15, 21)
64. The sum of the first twenty terms of the AP: $1\frac{1}{2}, 1\frac{1}{4}, 1, \dots$ is (a) -20 (b) -11 (c) -24.5 (d) -17.5
65. The sum of the first n terms of an AP is 252. If the first term is -16 and the last term is 72, find the number of terms n . (a) 9 (b) 10
(c) 6 (d) 7
66. Given an exponential sequence having 81 and 1 as the 1st and 5th terms respectively. Find the 7th term of the sequence. (a) $\frac{27}{4}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{1}{27}$
67. Evaluate the sum to infinity of the series: $1 + t + t^2 + t^3 + \dots$ (a) t^∞ (b) $\frac{9}{t^9}$ (c) $\frac{1}{1-t}$ (d) undefined
68. The n th term of the sequence 2, 6, 12, 20, 30, ... is (a) $n^2 - n$ (b) $\frac{1}{2}n + 1$ (c) $2n + \frac{n^2}{2}$ (d) $n(n + 1)$
69. A sequence of numbers $T_1, T_2, T_3, \dots, T_n$ satisfies the relation $T_{n+1} + n^2 = nT_n + 2$ for all integers $n \geq 1$. If $T_1 = 2$, Find
 $T = (T_2, T_3)$. (a) (1, 2) (b) (2, 3) (c) (3, 4) (d) (1, 3)
70. Find the roots of the equation: $x^3 - 2x^2 + x + 2 = 0$ (a) (1, 2, 3) (b) (-1, 2, 3) (c) (1, -2, 2) (d) (-1, -1, 2)

Paper 2 of 4 — MTH 101 Elementary or General Mathematics 2020 2021 First Semester

Level 100 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2020/2021

Course code & title: MTH 111 (General Mathematics 1) Duration: 2 % hrs

TYPE D

Instruction: Answer all questions. Submit your question paper and your answer sheet at the end of Examination.

1. Partial fractions of $\frac{1}{(x-a)(x-b)(x-c)}$ will be of form (a) $\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$ (b) $\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$ (c) $\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$ (d) $\frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$

(x-a)(x-b)(x-c) $\frac{1}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$

Br+cC

(x-b)(x-c)

2. Absolute value of the following integers -5,-1,0,2,4 in descending order can be best expressed as (a)5,4,2,1,0 (b)

0,1,2,4,5 (c)2,4,0,-5,-1 (d) 0,-1,2,-5,4

3. The distance between -55 and 10 on a real number line is (a)45 (b) -45 (c)-550 (d) 65

4. Let a, b, and c be real numbers. Which of the following is false (a) $a + (b + c) = (a + b) + c$ (b)

$a(b+c) = ab + ac$ (c) $a(b-c) =$

$ab - ac$ (d) $a(bc) = (ab)c$

5. Which of the following is not a subset of the set of real numbers (a) Natural numbers (b) Whole numbers (c) Irrational

numbers (d) complex numbers

6. Which of the following numbers is farthest from the number 1 in the number line (a)-11 (b) -5 (c) 5 (d) 11

7. Express 35° in radian measure (a) rad (b) $\frac{\pi}{6}$ rad (c) $\frac{\pi}{3}$ rad (d) rad

8. Find the area of a triangle with sides 10cm, 7cm and 5cm, (a) 17.25cm^2 (b) 16.25cm^2

(c) 18.25cm^2 (d) 15.25cm^2

9. Evaluate $\tan 75^\circ$ without the use of a calculator. (a) $2 + \sqrt{3}$ (b) $2 + \sqrt{2}$ (c) $3 + \sqrt{2}$ (d) $3 + \sqrt{3}$

10. If $\tan 26^\circ = 5$, find $\tan 8^\circ$. (a) $\frac{1}{5}$ (b) $\frac{1}{25}$ (c) $\frac{1}{125}$ (d) $\frac{1}{625}$

5 5 5

3 7 11

11. Given the sequence $5, 10, 15, \dots$: Find its nth term. (a) $5n$ (b) $2n-1$ (c) $5n-2$ (d) $-(2n-5)$

12. The third term of an arithmetic series is -5 and the fifth term is 4.5. Find the 13th term. (a) 33 (b) 42.5 (c) 14.5 (d) 47

13. If 19, x, y, -8 is an AP. What is (x,y)? (a) (12,7) (b) (7,12) (c) (10,1) (d) (15, 21)

14. The sum of the first twenty terms of the AP: 1, -1, 1, -1, ... is (a)-20 (b) -11 (c)-24.5 (d)-12.5

15. The sum of the first n terms of an AP is 252. If the first term is -16 and the last term is 72, find the number of terms n .
 (a) 9 (b) 10 (c) 6 (d) 7
16. Given an exponential sequence having 81 and 1 as the 1st and 5th terms respectively. Find the 7th term of the sequence.
 27
 (a) = (b) = (c) = (d) =
17. Evaluate the sum to infinity of the series: $1 + t + t^2 + t^3 + \dots$ (a) $\frac{1}{1-t}$ (b) $\frac{1}{1+t}$ (c) $\frac{1}{1-t^2}$ (d) undefined
18. The n th term of the sequence 2, 6, 12, 20, 30, ... is (a) n^2 (b) $n(n+1)$ (c) $2n^2$ (d) $n(n+1)$
19. A sequence of numbers $T_1, T_2, T_3, \dots$ satisfies the relation $T_{n+1} + n^2 = nT_n + 2$ for all integers $n > 1$. If $T_1 = 2$,
 Find $T = (T_2, T_3)$. (a) (1, 2) (b) (2, 3) (c) (2, 4) (d) (1, 3)
20. Find the roots of the equation: $x^3 - 2x^2 + x + 2 = 0$ (a) (1, 2, 3) (b) (-1, 2, 3) (c) (1, -2, 2) (d) (-1, 1, 2)
21. The quadratic equation whose roots are $2 - \sqrt{2}$ and $2 + \sqrt{2}$ is (a) $x^2 + x - 2 = 0$ (b) $5x^2 - 3x + 11 = 0$ (c) $2x^2 - 3x + 7 = 0$ (d) $x^2 - 4x + 2 = 0$
22. Solve the equations: $m^2 - n^2 = 9$, $m - n = 3$ (a) (1, 2) (b) (3, 0) (c) (-1, 3) (d) (2, 5)
23. Solve for y in the equation, $\log y + 5 = 5 - \log y$. (a) 1 (b) 2 (c) 3 (d) 4
24. For what value of k does the equation $3x^2 - 6x + k = 0$ have equal roots (a) 1 (b) 2 (c) 3 (d) 4
25. Find the value(s) of the number n such that twice its square is equal to three times the number plus 5. (a) (-1, 5) (b) (-2, 5) (c) (-1, 2) (d) (-2, 5)
26. Given that $U = \{x; -10 \leq x < 10\}$, $P = \{x; -10 < x < 10\}$ and $Q = \{x; -5 < x \leq 3\}$, which of the following is correct
 $IP \cap Q = OU \cup Q$ (a) $I \cap Q$ (b) $I \cap Q$ (c) $I \cap Q$ (d) $U \cap Q$
27. $|3x + 5| = 8$. Find all values of x which make this equation true (a) $\{1, -4\}$ (b) $\{4, -1\}$ (c) $\{1, -1\}$ (d) $\{4, -4\}$
28. If $A = \{x; x \leq 3\}$ and $B = \{x; x \geq 4\}$ find $A \cap B$ in interval form (a) $- \infty < x < 3$ (b) $- \infty < x < 3$ (c) $- \infty < x < 3$ (d) $- \infty < x < 3$
29. If $\cos \theta = \frac{1}{2}$ find $\sin \theta$. (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$
30. If $\cos \theta = \frac{1}{2}$; Find $\tan \theta$. (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{2}$
36. Express $\sin 60^\circ$ in terms of $\cos$. (a) $\frac{\sqrt{3}}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{2}$

37. Find the area of the sector of a circle radius 8cm having an angle = (a) rad (b) rad (c) rad (d) —*rad

V3x V8x/39 3 v3 v6 2

38. Evaluat Viixdz (a) = (b) — (c) > (d) —

39. Evaluate $2\sqrt{8} - \frac{3}{32} 4\sqrt{50}$ (a) 122 (b) $3\sqrt{90}$ (c) $6\sqrt{40}$ (a) $\sqrt{00}$

40. Let $M = \{\text{all positi}$

ve eve

n integers less than or equal to 20}, $Q = \{\text{all factors of 20}\}$. What is $R \cap Q$? (a) $\{1, 2, 4, 5, 10, 20\}$ (b) $\{2, 4, 10, 20\}$ (c) $\{6, 8, 10, 2, 14\}$ (d) $\{1, 4, 10, 0\}$

41. Find the sum of y° ,

42. Which of the following j

$A \cup (B \cap C)$ (b) $(A \cup B) \cap C$ (c) $A \cap (B \cup C)$ (d) none of the above

(ANC) (d) none of the above

43. Given the universal set $U = \{1, 2, 3, \dots, 10\}$, $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 5, 7, 9, 10\}$. Find $A \cap (B \cap C)$ (a) $\{3, 5, 7\}$ (b)

$\{3, 5, 7, 9, 10\}$ (c) $\{1, 2, 3, \dots, 10\}$ (d) $\{2, 4, 5, 6, 7, 8, 9, 10\}$

, 1 A p AB A Bx+c

44. Partial fractions Be rr will be of the form (a) — — Tz (b) orieied x41 y2=a, (G) none of the above

45. The nonnegative integer js equivalent to (a). $\{x: x > 0\}$ (b). $\{x \in \mathbb{Z}: 0 < x < \infty\}$ (c). $2^* \cup \{0\}$ (d). none of the above

46. If $X = \{x \in \mathbb{R}: x > -1\}$ and $Y = \{y \in \mathbb{R}: y \leq 1\}$. Find $(X \cap Y)'$. (a) $\{-1\}$ (b) $\emptyset$ (c) $\mathbb{R} - \{-1\}$ (d) $\mathbb{R} \cup \{-1\}$

1 1 4 1 1 1 1 oe. 3

47. $G \otimes 3 \setminus a - d$ IS equal to (aj> - 7 io) — = 'a= d) cere

. : x+5 B Bx+C A Bx+C

48. Partial fractions of ea are of the form (a) — + Z,, (b) = + rr (c) ae = ; (d) none of the above

49. Partial fraction of —

72 'S equivalent to (a) $Ax + \dots$ (b) — + — (j— (j=

90. Given $P = \{1, 3, 5, 7, 9, 11\}$ and $Q = \{2, 4, 6$

(d) 1

a product of two trig. Functions (a) $\cos 3x^\circ$

sag | Lee, ee) ee

\$33, (9 the form $a + bi$ (a) "Fe +h (b)

4. Find the value of $i^* + i^3 - ; 2 + i-$

95. Solve $2x+1 < yx - 2$, write the so]

96. Given $ay^* + by + c = 0$ jf p?

53. Express

the roots are ? (a) distinct (b) equal (c) complex (d) undefined

°7. If a and £ are the roots of the equation $3x^* + 5x - 2 =$ Find the value of — + r (a) - (b) - (c) : (d) -

oe 4 3y\3 27-1... 81, 27. gy

98. What is the coefficient of the 3 term in the €xpansion of $(2x \text{ os } >) ?$ (a) ~ (b) — (c) ma (d) —

29. Find the coefficient of y^* in the binomial expansion of $(1 - 2y)^n$

90. Find the value of n if the ratio of the coefficient of y^4 to that of y^j is $\frac{9}{7}$ (c) $\frac{7}{5}$ (d) $\frac{5}{4}$

21. Using Pascal triangle evaluate $(1 + \sqrt{3})^6 - (1 - \sqrt{3})^6$. (a) $9\sqrt{3}$ (b) $27\sqrt{3}$ (c) $32\sqrt{3}$ (d) $16\sqrt{3}$

2. The n th term of the series a^n is (a) a^n (b) a^n (c) a^n (d) a^n

$n - 1$ is an integer multiple of 5 is (a) 2 (b) 3 (c) 5 (d) 7

coefficient of x^7 in the expansion of $(1 + x)^{12}$ is (a) - (b) - (c) = (d)

of a series, find the 4th

3. The value of n when $5^{2n} + 3$

4. The coefficient

term. (a) 10 (b) 15 (c) 25 (d) 20

from A is (a) 3 (b) 4 (c) 8 (d) 16

7. The cardinality of $B = \{5, 6, 3, 2, 3, 2\}$ is (a) 6 (b) 5 (c) 4 (d) 3

3. If Z represents the set of integers, then $-3 < y < 2$ is (a) $\{-3, 2\}$ (b) $\{0, 1\}$ (c) $\{-3, -2, -1, 0, 1, 2\}$ (d) $\{$

. In a survey of 50 newspapers, 40 read Champ

Paper 3 of 4 — MTH 101 Elementary or General Mathematics 2018 2019 First Semester

Level 100 • First Semester • PastQuestion

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS 'COMPUTER SCIENCE

FIRST SEMESTER EXAMINATION 2018/2019

Course code & title: MTH 111 - General Mathematics |

Duration: 2% Hrs

TYPE H

NAME:

MATRIC NO:

DEPARTMENT: LEVEL:

2. What is the coefficient of the 3'

6. The coefficient of x^3 in the expansion of $(1 + x)^2$ is

7. Given the sequence $-5, 5, \dots$. Find its n th term.

8. The third term of an arithmetic series is -5 and the fifth term is 4.5 . Find the sum of the first 10 terms.

9. If $19, x, y, -8$ is an AP. What is (x, y) ?

10. The sum of the first n terms of an AP is 252 . If the first term is 3 , find the value of n .

Instruction: Answer all questions in this sheet using biro. Provide the correct answer to each question in the space provided.

10. The sum of the first n terms of an AP is 252 . If the first term is 3 , find the value of n .

1. The n th term of the series $1, 3, 5, \dots$ is $2n - 1$. Find the sum of the first n terms.

3

"term in the expansion of $(2x + 3)^{10}$ is $1023x^9$ "

3. Find the coefficient of y^* in the binomial expansion of $(x + y)^{10}$.

10, 2, 4 in descending order can be best described by the sequence $2, 6, 12, 20, 30, \dots$ is

$+ 3n - 1$ is an integer. Find the value of n .

11. The n th term of the sequence $2, 6, 12, 20, 30, \dots$ is

4. Absolute value of the following integers $-5, -10, -15, \dots$

5. The value of n when $52^n = 1024$

First term is -16 and the last term is 72 . Find the number of terms n .

12. Find the roots of the equation: $x^3 - 2x^2 + 4x - 4 = 0$

13. For what value of k does the equation $3x^2 - 6x + k = 0$ have equal roots?

14. If a and b are the roots of the equation $3x^2 + 5x - 2 = 0$. Find the value of $a^2 + b^2$.

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= 5,2 , ber of subs hich can be formed from A is 4

29. If set A @ {2,9,5,2}, then the number of subsets which can be fc i

30. The cardinality of B= (5,6, 3,2,3, 2} is 000

15. Evaluate $\tan 22.5^\circ$ without the use of a calculator,

Use the following to answer 17 and 18. Given that $\cos @ = +$

17. Find $\sin @$.

18. Find $\tan @$

19. $1 + \tan @$ is equivalent to

20. Find the area of a triangle with sides 10cm, 7em and Scm.

21. Evaluate $\tan 75^\circ$ without the use of a calculator,

22. If $\tan 2@ = 5$, find $\tan @$.

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23. Express 7 radian in degrees.

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24. Find the area of the sector of a circle radius 8em having an angle :

$V3 \times VB \times V39$

5

Use the following to answer 26, 27 and 28. Given that $\sin a = -$

13°

27. Evaluate $\cos a + \tan a$

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25. Evaluate

26. Evaluate $\cos'a$

. 3

28. Lvaluate $\operatorname{cosec}'a$

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ee ee ee 2 en a

31. It Z represents the set of integers, then $\{y: +3 < y < 2\}$ is

32. AC B isreed.....

$4 - 5 ' \gg . ' "$

33. If given the sets $A^{\text{TM}} \{1,2,3\}$ and $Be \{2,1\}$, what ts the relationship between them'

34. The set difference of the set A with null set is

33. The relation $(A \cup B)$ is equal to ...

36, Given the universal set $U = \{1, 2, 3, \dots, 10\}$, $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8\}$ and $C = \{3, 5, 7, 9, 10\}$. Find $A \cup (B \cap C)$...

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Paper 4 of 4 — MTH 101 PQ 2023

Level 100 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUM, DELTA STATE.
COLLEGE OF SCIENCE.
DEPARTMENT OF MATHEMATICS.
SECOND SEMESTER, 2022/2023 SESSION, TIME ALLOWED: 1Hr 15mins
TYPE C Instruction: Answer all question

Use the information below to answer question 1 and 2

If the ellipse has its major axis along the y-axis such that $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- What is the equation of the tangent (A) $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = 1$ (B) $\frac{x_1x}{b^2y_1} + \frac{y_1y}{a^2x_1} = 1$ (C) $\frac{x_1x}{a^2} - \frac{y_1y}{b^2} = 1$ (D) $\frac{x_1x}{a^2} + \frac{y_1y}{b^2} = 1$
- Obtain the equation of the normal N (A) $\frac{y_1y}{b^2y_1} + \frac{x_1x}{a^2x_1} = 1$ (B) $\frac{x_1x}{b^2y_1} + \frac{y_1y}{a^2x_1} = 1$ (C) $\frac{y_1y}{b^2y_1} - \frac{x_1x}{a^2x_1} = 1$ (D) $\frac{y_1y}{b^2y_1} + \frac{x_1x}{a^2x_1} = 1$
- The general form of the equation of an ellipse is? (A) $Ax^2 + By^2 + Cx + Dy + E = 0$ (B) $Ax^3 + By^3 + E = 0$ (C) $Ax^2 + By^2 + 2Cx + 2Dy + E = 0$ (D) $Ax^2 + By^2 - 2Cx - 2Dy + E = 0$
- Given the line L: $Ax + Bx + C = 0$ is a tangent to a circle $x^2 + y^2 - 2hx - 2ky + C = 0$, if the distance from the center O of the circle to the line L is equal to the radius r of the circle and the radius r is always perpendicular to line L, this states the condition of? (A) Tangency between the line L and the circle (B) normal between the line L and the circle (C) equation between the line L and the circle (D) point of intersection between the line L and the circle.
- The equation of a parabola is simply a Equation? (A) Quadratic (B) cubi (C) Quartic (D) linear.
- The segment that is formed when a line passes through the focus F of a parabola to the curve is called (A) Directrix (B) Latus Rectum (C) Eccentricity (D) Major Axis.
- The focal diameter of a latus rectum along the x-axis is calculated as (A) 3a (B) 4a (C) 5a (D) 6a
- The conic section with its eccentricity greater than one (1) is (A) Circle (B) Hyperbola (C) Ellipse (D) Parabola.
- Given that the equation of the parabola is $x^2 + 4x - 4y + 8 = 0$ and its axis of symmetry is parallel to the y-axis with y having degree 1, by comparison with the standard form, What is the value of a (i.e the perpendicular distance to the vertex)? (A) 2 (B) 1 (C) 4 (D) 8

Use the information provided below to answer question 10 and 11

Given that the equation of the ellipse is $4x^2 + 5y^2 - 24 = 0$

- Transform the equation of the ellipse above into its standard form. (A) $\frac{2x^2}{12} + \frac{4y^2}{6} = 0$ (B) $\frac{x^2}{12} + \frac{4y^2}{3} = 0$ (C) $\frac{x^2}{6} + \frac{y^2}{\frac{3}{2}} = 0$ (D) $\frac{x^2}{7} + \frac{2y^2}{24} = 0$
- What is the equation of the tangent at the point (1, 2)? (A) $4x + 10y - 16 = 0$ (B) $x + 5y - 8 = 0$ (C) $2x + 5y - 12 = 0$ (D) $8x + 8y - 24 = 0$
- What is the value of the latus rectum $\frac{2b^2}{a}$ given that the conic section is; $\frac{(x-1)^2}{16} - \frac{(y-2)^2}{16} = 1$
(A) 4 (B) 5 (C) 8 (D) 2.
- What is the distance between the point A(5, -2) and B(2, -7)? (A) $\sqrt{13}$ unit (B) $\sqrt{30}$ unit (C) $\sqrt{38}$ unit (D) $\sqrt{34}$ unit
- Find the shape of the line L which passes through the point (10, 3) and (4, 6). (A) 2 (B) $\frac{1}{3}$ (C) $-\frac{3}{2}$ (D) $-\frac{1}{2}$
- What is the equation of the line which passes through the point P(0, 1) and Q(-2, 0). (A) $2x + 2y + 2 = 0$ (B) $4x - 2y + 4 = 0$ (C) $8x - 4y - 12 = 0$ (D) $x - 2y + 2 = 0$
- Which of the following is correct? (A) $A \cdot B = |A||B|\cos\theta$ (B) $A \cdot B = |A||B|\sin\theta$ (C) $A \cdot B = |A||B|\tan\theta$ (D) $A \cdot B = |A||B|\cot\theta$
- Let $B = i + j$. Find the unit vector of B. (A) $\frac{1}{\sqrt{2}}(i + 3j)$ (B) $\frac{1}{\sqrt{2}}(i - 3j)$ (C) $\frac{1}{\sqrt{2}}(i - j)$ (D) $\frac{1}{\sqrt{2}}(i + j)$
- Suppose $A = 5i + 2j - 4k$ and $B = 4i + 3j$. Find $|A| + |B|$ (A) $5 + 3\sqrt{5}$ (B) $5 - 3\sqrt{5}$ (C) $7 + 3\sqrt{5}$ (D) $5 + 2\sqrt{5}$
- Which of the following is a vector quantity? (A) Length (B) Area (C) Force (D) Mass
- Let A and B be two nonzero vectors, then A and B are said to be parallel if
(A) $A \times B = 0$ (B) $A \times B = -2$ (C) $A \times B = -1$ (D) $A \times B = 1$
- Given that $B = 8i + 15j$. Find $|B|$ (A) 16 (B) 17 (C) 18 (D) 19
- Let $A = 8i + 15j$. Find the unit vector of A. (A) $\frac{1}{17}(8i + 15j)$ (B) $\frac{1}{17}(8i + 15j)$ (C) $\frac{1}{18}(8i + 15j)$ (D) $\frac{1}{19}(8i + 15j)$
- Let A and B be two perpendicular vectors, then A.B is given as (A) 0 (B) 1 (C) -1 (D) 2
- Which of the following is correct? (A) $|A||B| = \frac{\sin\theta}{A \times B}$ (B) $|A||B| = \frac{\cos\theta}{A \times B}$ (C) $|A||B| = \frac{A \times B}{\sin\theta}$ (D) $|A||B| = \frac{A \times B}{\cos\theta}$
- Given that $A = (2, 4, 3)$ and $B = (1, 5, -2)$. Evaluate $A \times B$ (A) $-23i + 7j + 6k$ (B) $-22i + 6j - 3k$ (C) $21i + j - 7k$ (D) $20i - j + 6k$